

Source: Larson & Gray, 2018 (7Ed)

Chapter 6: Developing a Project Plan

Q4.1. Draw AOA and AON networks for a project, using following information. What activity(ies) is a burst activity? What activity(ies) is a merge activity?

- | | | | |
|--------------------------|--|---|---|
| a) A<B
B<C,D
C,D<E | b) A<B
B<C
C<D,E,F
D,E,F<G
G<H | c) A<B,C,D,E
C,D<F
B,F<G
E,G<H | d) A<B
B<C,D,E
C<D
D<F
C<G
E,F,G<H |
|--------------------------|--|---|---|

**Note: A<B means B is preceeded by A.*

Q4.2. From the following information, develop an AON project network. Complete the forward and backward pass, compute activity slack, and identify the critical path. How many days will the project take?

ID	Description	Predecessor	Time
A	Survey site	None	2
B	Excavate site	A	4
C	Install power lines	B	3
D	Install drainage	B	5
E	Pour foundation	C, D	3

Q4.3. You have signed a contract to build a garage for the Simpsons. You will receive a \$500 bonus for completing the project within 17 working days. The contract also contains a penalty clause in which you will lose \$100 for each day the project takes longer than 17 working days.

Draw a project network given the information below. Complete the forward and backward pass, compute the activity slack, and identify the critical path. Do you expect to receive a bonus or a penalty on this project?

ID	Description	Predecessor	Time (days)
A	Prepare site	None	2
B	Pour foundation	B	3
C	Erect frame	C	4
D	Roof	C	4
E	Windows	C	1
	Doors	C	1
G	Electrical	D, E, F, G	3
H	Rough-in frame	F, G	2
I	Door opener	H, I	1
J	Paint	J	2
K	Cleanup		1

Q4.4. K. Nelson, project manager of Print Software, Inc., wants you to prepare a project network; compute the early, late, and slack activity times; determine the planned project duration; and identify the critical path. His assistant has collected the following information for the Color Printer Drivers Software Project:

ID	Description	Predecessor	Time
A	External specifications	None	8
B	Review design features	A	2
C	Document new features	A	3
D	Write software	A	60
E	Program and test	B	40
F	Edit and publish notes	C	2
G	Review manual	D	2
H	Alpha site	E, F	20
I	Print manual	G	10
J	Beta site	H, I	10
K	Manufacture	J	12
L	Release and ship	K	3